Normalization vs. Denormalization

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What is Normalization?

DATA ORGANIZATION STRATEGY

Normalization involves organizing data to minimize redundancy and dependency, thus improving data integrity and ensuring consistent, accurate information across databases.

What is Denormalization?

IMPROVING READ PERFORMANCE

Denormalization involves **combining data** to enhance performance by introducing redundancy, which speeds up data retrieval for read-heavy applications.

KEY DIFFERENCES BETWEEN STRATEGIES

Normalization focuses on minimizing redundancy and **ensuring data integrity**. This method creates complex queries due to multiple related tables, leading to **slower read times**. However, it allows for easier updates because changes in one table automatically reflect in related tables, promoting consistency across the database.

DENORMALIZATION EXPLAINED

Denormalization introduces redundancy by combining data from multiple tables into a single structure, resulting in **simpler queries**. This strategy enhances read performance significantly, making it ideal for read-heavy applications. Nevertheless, it complicates updates since changes must be made in multiple places, increasing the risk of inconsistency.

DENORMALIZATION CHARACTERISTICS

- Adds redundancy to data storage
- Simplifies query structure for users
- Speeds up data retrieval processes
- Requires careful management to avoid excess duplication
- Can lead to increased storage costs

NORMALIZATION CHARACTERISTICS

- Minimizes data redundancy and improves integrity
- Creates more complex queries that can slow down performance
- Ensures consistency across data entries
- Facilitates easier updates and changes to data tables
- Reduces chances of anomalies in data

Summary and Takeaway

STRATEGY SELECTION

- Use normalization for **data accuracy** and consistency.
- Opt for denormalization for **performance** and speed in read-heavy systems.
- The right approach depends on your **application needs** and data usage patterns.